

HW 3

1)
1)

$$z^* = \underset{z}{\operatorname{argmin}} z^T C z + c^T z + k$$

s.t.
 $Dz + b = 0$

$$z^T = [x_0^T \dots x_N^T, u_0^T \dots u_{N-1}^T]$$

$$z^T C z + c^T z + k = (x_N - x_d)^T P (x_N - x_d) + \sum_{k=0}^{N-1} [(x_k - x_d)^T Q (x_k - x_d) + u_k^T R u_k]$$

$$\rightarrow \sum_{k=0}^{N-1} \underline{x_k^T Q x_k} - 2 \underline{x_d^T Q x_k} + \underline{x_d^T Q x_d} + \underline{u_k^T R u_k}$$

$$(x_N - x_d)^T P (x_N - x_d) = x_N^T P x_N - x_N^T P x_d - x_d^T P x_N + x_d^T P x_d$$

$$= x_N^T P x_N - 2 x_N^T P x_d + x_d^T P x_d$$

$$= x_N^T P x_N - 2 x_N^T P x_d + x_d^T P x_d + \sum_{k=0}^{N-1} \underline{x_k^T Q x_k} - 2 \underline{x_d^T Q x_k} + \underline{x_d^T Q x_d} + \underline{u_k^T R u_k}$$

$$= z^T \begin{bmatrix} 0 & \dots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \dots & 0 \end{bmatrix} z^T + z^T \begin{bmatrix} 0 \\ \vdots \\ -2P x_d \\ 0 \end{bmatrix} + x_d^T P x_d + z^T \begin{bmatrix} Q & \dots & Q \\ \vdots & \ddots & \vdots \\ Q & \dots & R \end{bmatrix} z + z^T \begin{bmatrix} -2Q x_d \\ \vdots \\ -2Q x_d \\ 0 \end{bmatrix} + (x_d^T Q x_d) \cdot N$$

$\uparrow$ UP to $N-1$ th index

$$C = \begin{bmatrix} Q & \dots & Q \\ \vdots & \ddots & \vdots \\ Q & \dots & R \end{bmatrix}$$

$$\tilde{c} = \begin{bmatrix} -2Q x_d \\ \vdots \\ -2Q x_d \\ 0 \end{bmatrix} \leftarrow d^{\text{th}} \text{ index}$$

$$k = x_d^T P x_d + N(x_d^T Q x_d)$$

$$C = \begin{bmatrix} Q & \dots & Q \\ \vdots & \ddots & \vdots \\ Q & \dots & R \end{bmatrix}$$

Since Q , P and R are PSD, then the eigenvectors must all be larger or equal to 0. Thus, since C is a block diagonal matrix, it is also PSD.

□

2)

$$x_k = A^k x_0 + \sum_{i=0}^{k-1} A^{k-i-1} B u_i$$

Base case:

$$x_1 = A x_0 + B u_0 \quad \checkmark$$

Assume this holds for all $k = 1 \dots n$:

$n+1$:

$$\begin{aligned} x_{n+1} &= A x_n + B u_n \quad \swarrow \text{from } x_n \\ &= A \left(A^n x_0 + \sum_{i=0}^{n-1} A^{n-i-1} B u_i \right) + B u_n \\ &= A^{n+1} x_0 + \sum_{i=0}^{n-1} A^{n-i} B u_i + B u_n \\ &= A^{n+1} x_0 + \sum_{i=0}^n A^{n-i} B u_i \\ &= A^{n+1} x_0 + \sum_{i=0}^{(n+1)-1} A^{(n+1)-i-1} B u_i \quad \checkmark \end{aligned}$$

Thus, by induction, this holds $\square$

3)

$$Dz + b = 0$$

$$x_1 = Ax_0 + Bu_0$$

$$x_2 = A^2x_0 + ABu_0 + Bu_1$$

$$\vdots$$

$$x_N = A^Nx_0 + A^{N-1}Bu_0 + \dots + Bu_{N-1}$$

$$\begin{bmatrix} x_1 \\ \vdots \\ x_N \end{bmatrix} = \begin{bmatrix} A \\ A^2 \\ \vdots \\ A^N \end{bmatrix} x_0 + \begin{bmatrix} B & 0 & 0 & \dots & 0 \\ AB & B & 0 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ A^{N-1}B & A^{N-2}B & \dots & B \end{bmatrix} \begin{bmatrix} u_0 \\ \vdots \\ u_{N-1} \end{bmatrix}$$

$$Dz = -b$$

$$0 = \begin{bmatrix} A \\ A^2 \\ \vdots \\ A^N \end{bmatrix} x_0 + \begin{bmatrix} B & 0 & 0 & \dots & 0 \\ AB & B & 0 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ A^{N-1}B & A^{N-2}B & \dots & B \end{bmatrix} \begin{bmatrix} u_0 \\ \vdots \\ u_{N-1} \end{bmatrix} - \begin{bmatrix} x_1 \\ \vdots \\ x_N \end{bmatrix}$$

$$z = \begin{bmatrix} x_0 \\ X \\ U \end{bmatrix} \leftarrow \begin{matrix} X = \begin{bmatrix} x_1 \\ \vdots \\ x_N \end{bmatrix} \\ U = \begin{bmatrix} u_0 \\ \vdots \\ u_{N-1} \end{bmatrix} \end{matrix}$$

$$\text{Constraints: } x_0 = x(0)$$

$$Ax_0 - IX + BU = 0$$

$$\downarrow$$

$$\begin{bmatrix} I & 0 & 0 \\ A & -I & B \end{bmatrix} \begin{bmatrix} x_0 \\ X \\ U \end{bmatrix} = \begin{bmatrix} x(0) \\ 0 \end{bmatrix}$$

$$\Rightarrow D = \begin{bmatrix} I & 0 & 0 \\ A & -I & B \end{bmatrix}$$

$$b = \begin{bmatrix} -x(0) \\ 0 \end{bmatrix}$$

2,

1)

$$V(x_k) = \min_{\{u_i\}} \sum_{i=k}^{\infty} (x_i^T Q x_i + u_i^T R u_i)$$

$$= \min_u (x^T Q x + u^T R u + V(Ax + Bu))$$

$$V(x) = x^T P x \Rightarrow V(x_k) = \min_u (x^T Q x + u^T R u + (Ax + Bu)^T P (Ax + Bu))$$

$$x^T Q x + u^T R u + (Ax + Bu)^T P (Ax + Bu)$$

$$= x^T Q x + u^T R u + (u^T B^T + x^T A^T) P (Ax + Bu)$$

$$= x^T Q x + u^T R u + \underline{u^T B^T P A x} + u^T B^T P B u + x^T A^T P A x + \underline{x^T A^T P B u}$$

$$= x^T Q x + u^T R u + 2u^T B^T P A x + u^T B^T P B u + x^T A^T P A x$$

2)

$$J = x^T Q x + \underbrace{u^T R u + 2u^T B^T P A x}_{\text{quadratic in } u} + \underbrace{u^T B^T P B u + x^T A^T P A x}$$

$$= u^T (R + B^T P B) u + 2x^T A^T P B u + x^T A^T P A x + x^T Q x$$

$$\nabla_u J = (R^T + B^T P^T B + R + B^T P B) u + 2B^T P A x$$

$$= 0$$

$$\Rightarrow (R^T + B^T P^T B + R + B^T P B) u = -2B^T P A x$$

↓ symmetric

$$\Rightarrow 2(R + B^T P B) u = -2B^T P A x$$

$$\Rightarrow u^* = -(R + B^T P B)^{-1} B^T P A x$$

$$\text{so, } K = (R + B^T P B)^{-1} B^T P A$$

3)

$$\vec{u} = -(R+B^T P B)^{-1} B^T P A x$$

$$V(x) = \min_u (x^T Q x + u^T R u + V(Ax + Bu))$$

$$\begin{aligned} V^*(x) &= x^T Q x + (-kx)^T R (-kx) + (Ax - BKx)^T P (Ax - BKx) \\ &= x^T P x \end{aligned}$$

$$\rightarrow x^T Q x + (-kx)^T R (-kx) + (Ax - BKx)^T P (Ax - BKx) = x^T P x$$

$$\rightarrow x^T Q x + x^T K^T R K x + (-x^T K^T B^T + x^T A^T) P (Ax - BKx) = x^T P x$$

$$\rightarrow Q + K^T R K + (-K^T B^T + A^T) P (A - BK) = P$$

$$\rightarrow Q + K^T R K - K^T B^T P A + K^T B^T P B K + A^T P A - A^T P B K = P$$

$$\rightarrow Q + K^T R K - 2K^T B^T P A + K^T B^T P B K + A^T P A = P$$

$$\rightarrow Q + K^T (R + B^T P B) K - 2K^T B^T P A + A^T P A = P$$

$$\downarrow K = (R + B^T P B)^{-1} B^T P A$$

$$Q + A^T P^T B K - 2K^T B^T P A + A^T P A = P$$

$$\rightarrow Q - K^T B^T P A + A^T P A = P$$

$$\downarrow K = (R + B^T P B)^{-1} B^T P A$$

$$\rightarrow Q - A^T P^T B (R + B^T P B)^{-1} B^T P A + A^T P A = P$$

□

3,

1)

a)

$$P_{\theta}(\tau) = P(s_0) \prod_{t=0}^{\tau} \pi_{\theta}(a_t | s_t) P(s_{t+1} | s_t, a_t)$$

$$J(\theta) = \mathbb{E}_{\tau} [R(\tau)]$$

$$= \mathbb{E}_{\tau} \left[\sum_{t=0}^{\tau} r(s_t, a_t) \right]$$

$$= \int_{\tau} R(\tau) P_{\theta}(\tau) d\tau$$

$$\nabla_{\theta} J(\theta) = \nabla_{\theta} \int_{\tau} R(\tau) P_{\theta}(\tau) d\tau$$

$$= \int_{\tau} R(\tau) \nabla_{\theta} P_{\theta}(\tau) d\tau$$

b)

$$\nabla_{\theta} P_{\theta}(\tau) = P_{\theta}(\tau) \nabla_{\theta} \log(P_{\theta}(\tau))$$

$$\nabla_{\theta} J(\theta) = \int_{\tau} R(\tau) P_{\theta}(\tau) \nabla_{\theta} \log(P_{\theta}(\tau)) d\tau$$

$$= \mathbb{E}_{\tau} [R(\tau) \nabla_{\theta} \log(P_{\theta}(\tau))]$$

c)

$$\nabla_{\theta} J(\theta) = \mathbb{E}_{\tau} [R(\tau) \nabla_{\theta} \log(P_{\theta}(\tau))]$$

$$= \mathbb{E} \left[R(\tau) \nabla_{\theta} \log \left(P(s_0) \prod_{t=0}^{\tau} \pi_{\theta}(a_t | s_t) P(s_{t+1} | s_t, a_t) \right) \right]$$

$$= \mathbb{E} \left[R(\tau) \nabla_{\theta} \left(\log(s_0) + \sum_{t=0}^{\tau} \log(\pi_{\theta}(a_t | s_t)) + \log(s_{t+1} | s_t, a_t) \right) \right]$$

$$= \mathbb{E} \left[R(\tau) \nabla_{\theta} \sum_{t=0}^{\tau} \log(\pi_{\theta}(a_t | s_t)) \right] \quad \leftarrow \begin{array}{l} \text{don't} \\ \text{on } \theta \end{array} \text{ depend}$$

As desired



2)

a)

$$\nabla_{\theta} J(\theta) = \mathbb{E} \left[\sum_{t=0}^T \nabla_{\theta} \log(\pi_{\theta}(a_t | s_t)) (R(t) - b(s_t)) \right]$$

$$= \mathbb{E} \left[\sum_{t=0}^T \nabla_{\theta} \log(\pi_{\theta}(a_t | s_t)) R(t) - \sum_{t=0}^T \nabla_{\theta} \log(\pi_{\theta}(a_t | s_t)) b(s_t) \right]$$

$$= \mathbb{E} \left[\sum_{t=0}^T \nabla_{\theta} \log(\pi_{\theta}(a_t | s_t)) R(t) \right] - \mathbb{E} \left[\sum_{t=0}^T \nabla_{\theta} \log(\pi_{\theta}(a_t | s_t)) b(s_t) \right]$$

$$\mathbb{E} \left[\sum_{t=0}^T \nabla_{\theta} \log(\pi_{\theta}(a_t | s_t)) b(s_t) \right] = \mathbb{E}_{\zeta} \left[\mathbb{E}_{\alpha} \left[\sum_{t=0}^T \nabla_{\theta} \log(\pi_{\theta}(a_t | s_t)) b(s_t) \mid \zeta_t \right] \right]$$

$$= \sum_{t=0}^T \mathbb{E}_{\zeta} \left[b(s_t) \mathbb{E}_{\alpha} \left[\nabla_{\theta} \log(\pi_{\theta}(a_t | s_t)) \mid \zeta_t \right] \right]$$

$$= \sum_{t=0}^T \mathbb{E}_{\zeta} \left[b(s_t) \mathbb{E}_{\alpha} \left[\nabla_{\theta} \log(\pi_{\theta}(a_t | s_t)) \mid \zeta_t \right] \right]$$

$$= \sum_{t=0}^T \mathbb{E}_{\zeta} \left[b(s_t) \int_{\alpha} p_{\theta}(\alpha) \nabla_{\theta} \log(\pi_{\theta}(a_t | s_t)) \right]$$

$$= \sum_{t=0}^T \mathbb{E}_{\zeta} \left[b(s_t) \int_{\alpha} \pi_{\theta}(\alpha | s_t) \nabla_{\theta} \log(\pi_{\theta}(a_t | s_t)) \right]$$

$$= 0$$

$$\Rightarrow \nabla_{\theta} J(\theta) = \mathbb{E} \left[\sum_{t=0}^T \nabla_{\theta} \log(\pi_{\theta}(a_t | s_t)) R(t) \right] \quad \checkmark$$

b)

- i. $g = \nabla_{\theta} \log(\pi_{\theta}(a|s)) R$
 ii. $g_b = \nabla_{\theta} \log(\pi_{\theta}(a|s)) (R - b)$

iii.

$$E[g] = E[\nabla_{\theta} \log(\pi_{\theta}(a|s)) R]$$

$$E[g_b] = E[\nabla_{\theta} \log(\pi_{\theta}(a|s)) R] - E[\nabla_{\theta} \log(\pi_{\theta}(a|s)) b]$$

$$= E[\nabla_{\theta} \log(\pi_{\theta}(a|s)) R] - \sum_a \nabla_{\theta} \log(\pi_{\theta}(a|s)) b \pi_{\theta}(a|s)$$

$$= E[\nabla_{\theta} \log(\pi_{\theta}(a|s)) R] - b \sum_a \nabla_{\theta} \log(\pi_{\theta}(a|s)) \pi_{\theta}(a|s)$$

$$= E[\nabla_{\theta} \log(\pi_{\theta}(a|s)) R] - b \sum_a \nabla_{\theta} \log(\pi_{\theta}(a|s)) \pi_{\theta}(a|s)$$

Using hint from 2) a):

$$b \sum_a \nabla_{\theta} \log(\pi_{\theta}(a|s)) \pi_{\theta}(a|s) = 0$$

$$\Rightarrow E[g] = E[g_b]$$

□

iv.

$$\text{Var}(g_b) = E[g_b^2] - E[g_b]^2$$

$$\min_{b(s)} \text{Var}(g_b) \equiv \min_{b(s)} E[g_b^2]$$

$$E[g_b^2] = E[\nabla_{\theta}^2 \log(\pi_{\theta}(a|s)) (R - b(s))^2]$$

$$= E[\nabla_{\theta}^2 \log(\pi_{\theta}(a|s)) R^2] - 2b(s) E[\nabla_{\theta}^2 \log(\pi_{\theta}(a|s)) R] + b(s)^2 E[\nabla_{\theta}^2 \log(\pi_{\theta}(a|s))]$$

$$= E[\nabla_{\theta}^2 \log(\pi_{\theta}(a|s)) R^2] - 2b(s) E[\nabla_{\theta}^2 \log(\pi_{\theta}(a|s)) R] + b(s)^2 E[\nabla_{\theta}^2 \log(\pi_{\theta}(a|s))]$$

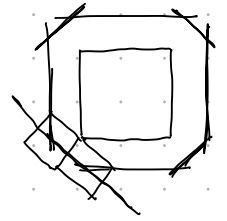
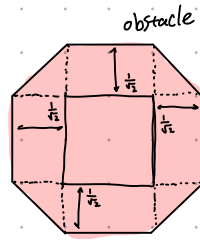
$$\frac{\partial}{\partial b(s)} E[g_b^2] = -2 E[\nabla_{\theta}^2 \log(\pi_{\theta}(a|s)) R] + 2b(s) E[\nabla_{\theta}^2 \log(\pi_{\theta}(a|s))] = 0$$

$$b(s) = \frac{E[\nabla_{\theta}^2 \log(\pi_{\theta}(a|s)) R]}{E[\nabla_{\theta}^2 \log(\pi_{\theta}(a|s))]}$$

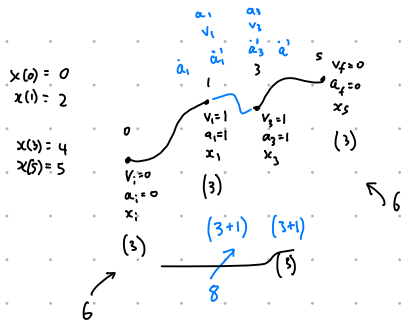
□

4,

1)



2)

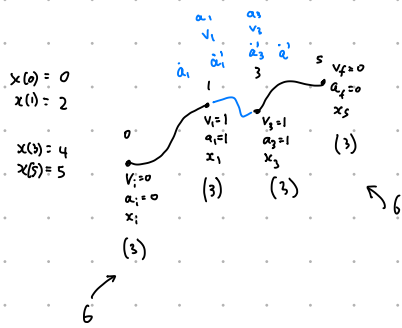


max # constraints is 8, so polynomial degree 7.

3)

Yes, since the starting and end velocity, position and acceleration are provided, so the segments don't depend on each other.

4)



$$C(x) = a_5x^5 + a_4x^4 + a_3x^3 + a_2x^2 + a_1x + a_0$$

$$\dot{C}(x) = 5a_5x^4 + 4a_4x^3 + 3a_3x^2 + 2a_2x + a_1$$

$$\ddot{C}(x) = 20a_5x^3 + 12a_4x^2 + 6a_3x + 2a_2$$

Curve 1:

$$\begin{bmatrix} x(0) \\ x(1) \\ \dot{x}(0) \\ \dot{x}(1) \\ \ddot{x}(0) \\ \ddot{x}(1) \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 5 & 4 & 3 & 2 & 1 & 0 \\ 0 & 0 & 0 & 2 & 0 & 0 \\ 20 & 12 & 6 & 2 & 0 & 0 \end{bmatrix} \begin{bmatrix} a_5 \\ a_4 \\ a_3 \\ a_2 \\ a_1 \\ a_0 \end{bmatrix} = \begin{bmatrix} 0 \\ 2 \\ 0 \\ 1 \\ 0 \\ 1 \end{bmatrix}$$

Curve 2:

$$\begin{bmatrix} x(1) \\ x(3) \\ \dot{x}(1) \\ \dot{x}(3) \\ \ddot{x}(1) \\ \ddot{x}(3) \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 \\ 243 & 81 & 27 & 9 & 3 & 1 \\ 5 & 4 & 3 & 2 & 1 & 0 \\ 405 & 108 & 27 & 6 & 1 & 0 \\ 20 & 12 & 6 & 2 & 0 & 0 \end{bmatrix} \begin{bmatrix} a_5 \\ a_4 \\ a_3 \\ a_2 \\ a_1 \\ a_0 \end{bmatrix} = \begin{bmatrix} 2 \\ 4 \\ 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

Curve 3

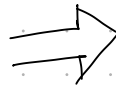
function M = quintic_matrix(t_i, t_f)

```
M = [t_i^5, t_i^4, t_i^3, t_i^2, t_i, 1;
      t_f^5, t_f^4, t_f^3, t_f^2, t_f, 1;
      5*t_i^4, 4*t_i^3, 3*t_i^2, 2*t_i, 1, 0;
      5*t_f^4, 4*t_f^3, 3*t_f^2, 2*t_f, 1, 0;
      20*t_i^3, 12*t_i^2, 6*t_i, 2, 0, 0;
      20*t_f^3, 12*t_f^2, 6*t_f, 2, 0, 0];
```

end

```
c_1 = quintic_matrix(0,1) \ [0;2;0;1;0;1]
c_2 = quintic_matrix(1,3) \ [2;4;1;1;1;1]
c_3 = quintic_matrix(3,5) \ [4;5;1;0;1;0]
```

Constraints



$$C_1 = \begin{bmatrix} 9.5 \\ -2.4 \\ 16.5 \\ 0 \\ 0 \\ 0 \end{bmatrix} \quad C_2 = \begin{bmatrix} 0 \\ 0.125 \\ -1 \\ 2.75 \\ -2 \\ 2.125 \end{bmatrix} \quad C_3 = \begin{bmatrix} -0.0625 \\ 1.375 \\ -11.8750 \\ 50 \\ -101.5625 \\ 83.125 \end{bmatrix}$$